MONAI-CARDIAC-FAT-FL • CLINICAL EDGE CASES, DISCORDANT PHENOTYPES & 3D VLM DISCOVERY

Deep Cohort Mining Across N = 1,001 Quantified Stanford COCA CTs & N = 996 NV-Reason-CT-4B 3D Radiologist Narratives

Demonstrating Subclinical Vascular Inflammation Detection (CAC = 0), Pathologic Confounder Attribution, and Automated DICOM Geometry Recovery

1,001 / 1,002 (99.9%) Quantified

996 3D NV-Reason-CT Narratives

CRISP-CT Cutoff: > -70.1 HU

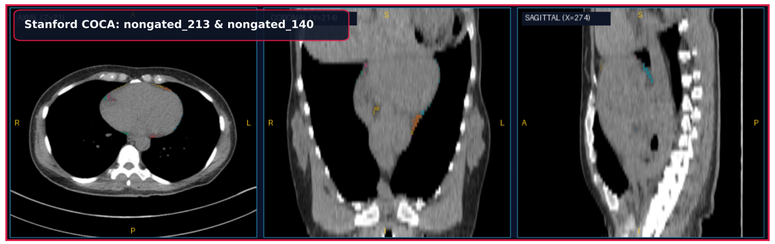
Zero-Spacing DICOM Recovery

100 Implanted Devices Flagged

60 Pericardial Effusions Caught

1. 'INVISIBLE RISK': CAC = 0 + SEVERE INFLAMMATION

Subclinical Coronary Microvascular Inflammation Missed by Agatston Scoring

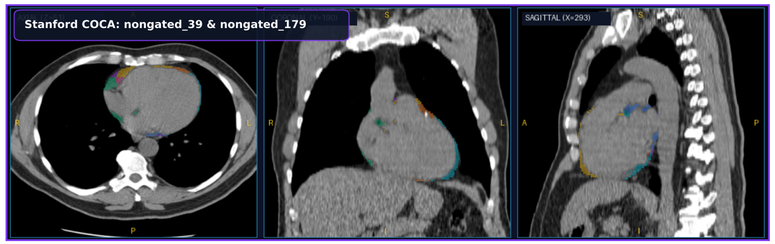


Agatston CAC: 0.0 (Zero Tier) | PVAT: -55.8 HU (Active Inflammation > -70.1)

- Standard calcium scoring classifies patient as Zero Risk (CAC = 0).
- **3D 10mm Pericoronary EDT reveals severe perivascular attenuation (-55.8 HU** in nongated_213; -51.1 HU in nongated_140; -60.3 HU in nongated_24).
- **NV-Reason-CT 3D VLM correlates perivascular edema & soft-plaque haze:** **"Pericoronary fat attenuation above -70.1 HU cutoff indicates active** coronary microvascular inflammation despite absence of calcified plaque."

2. PATHOLOGIC CONFOUNDERS: EFFUSION & HYDROPNEUMOTHORAX

3D Vision-Language Radiologist Explains Extreme Pericoronary HU Shifts

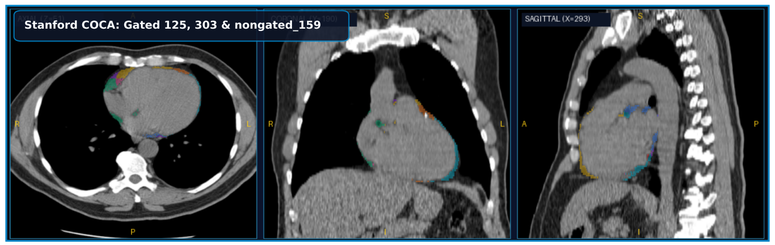


PVAT: -53.1 HU (nongated_39) & -47.7 HU (nongated_179) | CAC: 4,506 & 1,321

- Why is PVAT attenuation so elevated (-53.1 to -47.7 HU)?
- In nongated_39, NV-Reason-CT autonomously detects post-sternotomy **"Loculated hyperdense collection between the pericardial leaves** (hemorrhagic pericardial effusion?)" + "left fissure phantom tumor."
- In nongated_179 (-47.7 HU), NV-Reason-CT flags "Moderate-to-severe left hydropneumothorax with left lung volume loss & mediastinal shift."

3. EPICARDIAL LIPOMAS & STABILIZED MASSIVE PLAQUE

Distinguishing Benign Lipomatous Masses & Quiescent Heavy Calcification

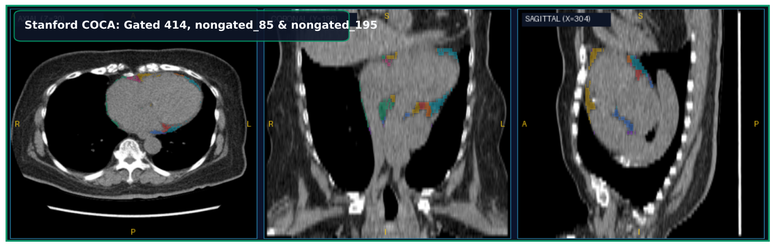


CAC: 635,362 (Case 125) & 619,849 (Case 303) | PVAT: -87.0 to -89.9 HU (Quiescent)

- High CAC does NOT always mean active vascular inflammation: Cases 125 & 303 have massive calcified burden (>600k) but quiescent PVAT (-89.9 HU).
- Strict -30.0 HU upper threshold prevents dense calcium (>130 HU) from blooming into adjacent pericoronary adipose tissue.
- NV-Reason-CT flags rare focal masses: "Hypodense fatty masses in the epicardial fat consistent with lipomas" (nongated_159 & Case 125).

4. BIOMARKER EXTREMES & CORRUPTED DICOM RECOVERY

From 327.6 cm³ Visceral Obesity to Cachexia & Zero-Spacing Header Rescue



Max EAT: 327.6 cm³ (Case 414) | Min PVAT: 0.00 cm³ (nongated_85) | dz=5.0mm Rescue

- Extreme Visceral Obesity (Case 414): EAT = 327.57 cm³ (3.3x cohort mean) with 46.24 cm³ PVAT, accurately bounded by DeepHeartSeg ONNX pericardium.
- Cachectic Empty-Groove Guardrail (nongated_85): EAT = 6.10 cm³, PVAT = 0.00 cm³ -> emits explicit NaN HU instead of false 0.0 HU water artifact.
- Corrupted Scanner Header Rescue (nongated_195): Scanner wrote SpacingBetweenSlices = 0.000000; auto-recovered (dz = 5.0mm) via IPP[2].

5. AUTONOMOUS 3D VLM DISCOVERY SPECTRUM (N = 996 CTs)

NVIDIA NV-Reason-CT-4B (3D ViT + Qwen3.5-4B) Structured Findings

Coronary Artery Wall Calcification	761 (76.4%)
Cardiomegaly / Chamber Enlargement	411 (41.3%)
Basilar / Dependent Atelectasis	168 (16.9%)
Pulmonary Fibrotic Sequelae / Scarring	149 (15.0%)
Incidental Pulmonary Nodule(s)	141 (14.2%)
Thoracic Aortic Wall Calcification	116 (11.6%)
Implanted Devices / Sternotomy / Valves	100 (10.0%)
Ascending Aorta / Main PA Dilation	76 (7.6%)
Pericardial Effusion / Thickening	60 (6.0%)
Mediastinal / Hilar Lymphadenopathy	54 (5.4%)
Esophageal Hiatal Hernia	34 (3.4%)
Pleural Effusion / Hydropneumothorax	25 (2.5%)
Bone Metastases / Rib Fractures / Masses	12 (1.2%)

HIGH-VALUE INCIDENTAL & DIAGNOSTIC DISCOVERIES FLAGGED BY 3D VLM

- **Case 442 (CAC = 4,169, PVAT = -67.6 HU): Coronary vasa vasorum neovascularization** infiltrating pericoronary space + incidental upper-abdominal ascites.
- **Case 276 (CAC = 53,938, EAT = 126.8 cm³): Left ventricular myocardial lipomatous** metaplasia (post-MI scar fat) + dilated main pulmonary artery (pulm. HTN).
- **Case nongated_151 (EAT = 104.2 cm³): Dual-chamber pacemaker (RA/RV leads) +** combined prosthetic aortic and mitral valve replacements.
- **Case nongated_24 (CAC = 0, PVAT = -60.3 HU): Multiple sclerotic thoracic osseous** metastases + large right pleural effusion on a zero-calcium scan.
- **Case nongated_51 & 632: Bilateral lower-lobe & right middle-lobe pulmonary masses** with ipsilateral hemidiaphragm elevation requiring urgent oncologic follow-up.

ENGINEERING RESILIENCE FOR CLINICAL EDGE CASES: 5 FAIL-SAFE PIPELINE GUARDRAILS

1. Zero-Spacing DICOM Recovery

Reconstructs 3D volume & physical dz from ImagePositionPatient[2] when SpacingBetweenSlices = '0.000000' *(rescued nongated_195, 213, 214).*

2. Multi-Series Scout Filter

Selects max(series_ids, key=len) + deduplicates identical Z-coordinate slices to prevent 1-slice topogram *collisions in hospital PACS archives.*

3. Strict -30.0 HU Blooming Cutoff

Enforces [-190.0, -30.0] HU window so dense coronary calcium (>130 HU), stents, and contrast lumen never *contaminate PVAT attenuation.*

4. Empty-Groove NaN Guardrail

Emits 0.0 cm³ volume and explicit NaN attenuation when a 10mm vessel cylinder has 0 fat voxels (prevents *false 0.0 HU water-density spike).*

5. Anisotropic 3D Physical EDT

Passes exact (dz, dy, dx) mm spacing to scipy distance_transform_edt so 10.0 mm PVAT dilation is invariant *across 0.5mm Gated & 5.0mm Non-Gated.*